Essential Summary

Quantitative Analysis of Dynamical Bifurcations in a Coupled Smooth and Discontinuous Oscillator with High-order Nonlinear Damping

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1 Background and Motivation

The coupled smooth and discontinuous (SD) oscillator, introduced by Han et al. (2012), consists of two SD oscillators coupled rigidly. Its restoring force contains **two irrational nonlinear terms** arising from the geometric configuration of oblique springs. These irrational terms make the system significantly harder to analyze than classical nonlinear oscillators.

When perturbed by high-order nonlinear damping of the form $(\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x}$, the oscillator exhibits rich dynamical behavior: limit cycles of varying stability, homoclinic and heteroclinic bifurcations, and multi-stable configurations (single-well, double-well, and triple-well potentials).

Classical quantitative methods—the elliptic function L–P method, the elliptic perturbation method, and the standard GHFP method—**cannot handle** this system because the two irrational terms render the key integrals analytically intractable. This work overcomes that barrier by extending the **modified generalized harmonic function perturbation (MGHFP)** method to this most challenging case.

2 System Model

The investigated oscillator is governed by:

$$\ddot{x} + (x+\beta)\frac{1}{\sqrt{(x+\beta)^2+\alpha^2}} + (x-\beta)\frac{1}{\sqrt{(x-\beta)^2+\alpha^2}} = \varepsilon(\mu_c + \mu_1 x + \mu_2 x^2 + \mu_3 x^3 + \mu_4 x^4)\dot{x} \quad (1)$$

where α is the smoothing parameter ($\alpha \to 0$ gives the non-smooth/discontinuous limit), β is the coupling parameter controlling the potential well structure, ε is a small positive perturbation parameter, μ_c is the control parameter, and μ_1–μ_4 are nonlinear damping coefficients.

2.1 Triple-Well Potential Structure

As α and β vary, the oscillator's potential energy landscape undergoes qualitative changes. For the triple-well configuration, the system has **five equilibrium points**: two centers, two saddles, and an additional center—creating a landscape where both homoclinic and heteroclinic orbits can coexist at the same saddle point.

3 The MGHFP Method — Key Steps

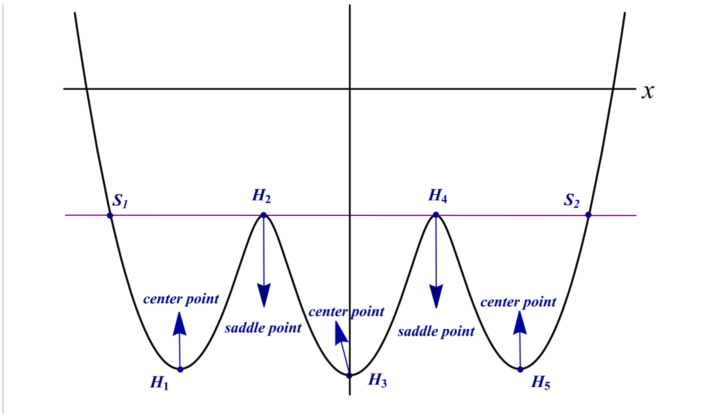

Figure 1: **Figure 5.** Potential energy diagram of the oscillator in the triple-well configuration. The five equilibrium points create a complex landscape with three potential wells separated by saddle points, enabling coexistence of homoclinic and heteroclinic orbits.

3.1 Step 1: Nonlinear Time Transformation

Introduce $\frac{d\varphi}{dt} = \Phi(\varphi)$ with $\varphi(0) = 0$, $\varphi(t + \tau) = \varphi(t) + \pi$. The undamped system transforms into:

$$\frac{d}{d\varphi}(\Phi x') + g(x) = 0, \qquad x(\varphi) = a\cos^2\Phi(\varphi) + b \tag{2}$$

where $g(x) = (x + \beta)\left[1 - \frac{\alpha^2}{(x+\beta)^2+\alpha^2}\right] + (x - \beta)\left[1 - \frac{\alpha^2}{(x-\beta)^2+\alpha^2}\right]$.

The function $\Phi_0(\varphi)$ is expanded as a Fourier series:

$$\Phi_0(\varphi) = \sum_{i=0}^{m} (p_{2i}\cos 2i\varphi + q_{2i}\sin 2i\varphi) \tag{3}$$

3.2 Step 2: Perturbation Expansion

The amplitude and frequency are expanded in powers of ε:

$$a = a_0 + \varepsilon a_1 + \cdots, \qquad \Phi = \Phi_0 + \varepsilon\Phi_1 + \cdots \tag{4}$$

Substituting into the damped system and collecting terms at each order of ε yields a hierarchy of equations. The zeroth-order gives the unperturbed solution; the first-order determines a_1 and Φ_1.

3.3 Step 3: Composite Simpson Integration (Key Innovation)

The Fourier coefficients p_{2i} involve integrals of the irrational nonlinear terms that **cannot be evaluated analytically**. The MGHFP method's key innovation is the introduction of the **composite Simpson quadrature formula** to discretize these integrals into sums over equally-spaced nodes. The integration interval $[0, \pi]$ is divided into eight equal parts, yielding explicit expressions for each p_{2i} in terms of a_0, b, α, and β.

4 Key Equations

4.1 Amplitude–Parameter Relation (Central Result)

Setting the Fourier truncation order $m = 4$, the analytical relationship between the limit cycle amplitude A and the control parameter μ_c is:

$$\boxed{\mu_c = -\frac{4}{a_0^2(2p_0 - p_4)}\Big(E_1\mu_1 + E_2\mu_2 + E_3\mu_3 + E_4\mu_4\Big)} \tag{5}$$

where p_{2i} are Fourier coefficients computed via composite Simpson integration, and E_1–E_4 are explicit functions of a_0, b, and p_{2i}. For symmetric limit cycles: $a_0 = 2A$, $b = -A$. For asymmetric limit cycles: $a_0 = A - C$, $b = C$, where C depends on α and β.

4.2 Stability Criterion

The stability of each limit cycle is determined by the characteristic quantity:

$$\boxed{h_0 = \frac{1}{\pi}\int_0^{\pi}\left[\mu_c + \sum_{i=1}^{4}\mu_i(a_0\cos^2\varphi + b)^i\right]d\varphi} \tag{6}$$

Based on the qualitative theory of ODEs:

- $h_0 > 0$: the limit cycle is **unstable**
- $h_0 = 0$: the limit cycle is **semi-stable** (bifurcation point)
- $h_0 < 0$: the limit cycle is **stable**

4.3 Homo-heteroclinic Bifurcation Thresholds

Setting $b = h$ (saddle point ordinate) in Eq. (5) yields μ_c^{hom} for **homoclinic** bifurcation. Setting $b = h_2$ and $a_0 + b = h_1$ (connecting two saddles) yields μ_c^{het} for **heteroclinic** bifurcation.

4.4 First-Order Approximate Solution

$$x = 2A\cos^2\Phi - A + \mathcal{O}(\varepsilon^2), \qquad \dot{x} = -2a_0[\Phi_0 + \varepsilon\Phi_1]\cos\varphi\sin\varphi + \mathcal{O}(\varepsilon^2) \tag{7}$$

5 Example 1: Single-Well Potential ($\alpha = 2$, $\beta = 0.5$)

With $\mu_1 = 1$, $\mu_2 = 1$, $\mu_3 = 1$, $\mu_4 = -1$, the amplitude–parameter curve reveals the complete limit cycle lifecycle:

The lifecycle unfolds as follows:

- $\mu_c < \mu^{(0)} = -0.1249$: **no limit cycle exists**.
- At $\mu_c = \mu^{(0)}$: a **semi-stable** limit cycle emerges (outer-stable, inner-unstable).
- $\mu^{(0)} < \mu_c < 0$: the semi-stable cycle **splits into two**—a smaller unstable cycle ($h_0 > 0$) and a larger stable cycle ($h_0 < 0$).
- At $\mu_c = 0$: the unstable cycle **collapses** to the singular point.
- $\mu_c > 0$: only a single **stable** limit cycle persists, growing with μ_c.

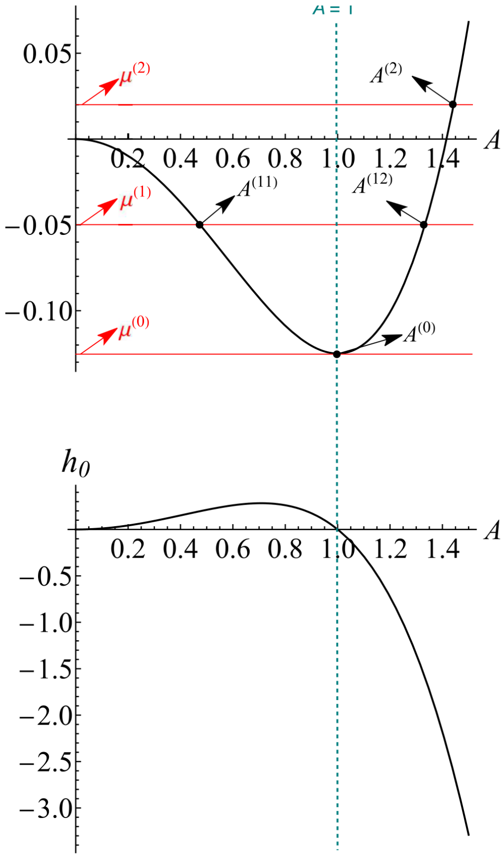

Figure 2: **Figure 1.** (Upper) Relationship between limit cycle amplitude A and control parameter μ_c. (Lower) Characteristic quantity h_0 determining stability. Key bifurcation points: $\mu^{(0)} = -0.1249$ (semi-stable cycle emerges), $\mu^{(1)} = -0.05$ (splits into stable + unstable), $\mu^{(2)} = 0.02$ (unstable cycle collapses to singular point).

6 Example 2: Triple-Well Potential

The triple-well case is considerably more complex: five equilibrium points, coexistence of homoclinic and heteroclinic orbits, and both small and large limit cycles.

6.1 Amplitude–Parameter Relation for Triple-Well

6.2 Limit Cycle Phase Portraits

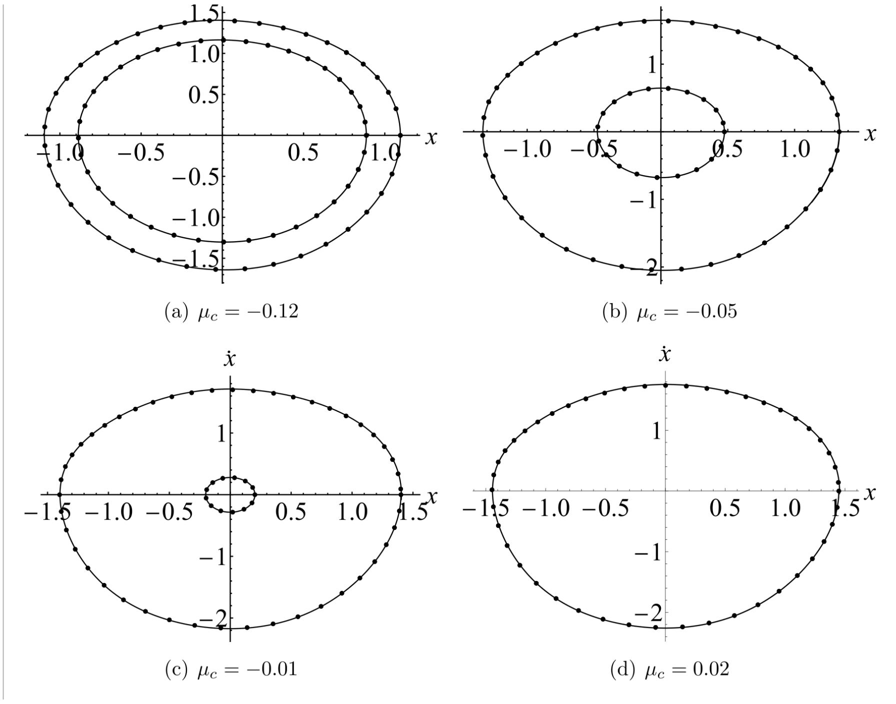

Figure 3: **Figure 2.** Phase portraits of limit cycles at different μ_c values ($\varepsilon = 0.2$). Solid lines: Runge–Kutta numerical integration. Dotted lines: present MGHFP analytical method. The excellent agreement validates accuracy even at moderately large ε.

6.3 Homoclinic Orbits

The homoclinic orbits connect a saddle point to itself. The MGHFP method provides analytical approximate solutions for these orbits, which are validated against numerical integration.

6.4 Heteroclinic Orbits

The heteroclinic orbits connect two distinct saddle points. For the triple-well system, these coexist with homoclinic orbits at the same saddle—a uniquely challenging scenario that the MGHFP method successfully handles.

Figure 5. Potential energy diagram of the oscillator (66) in triple well potential.

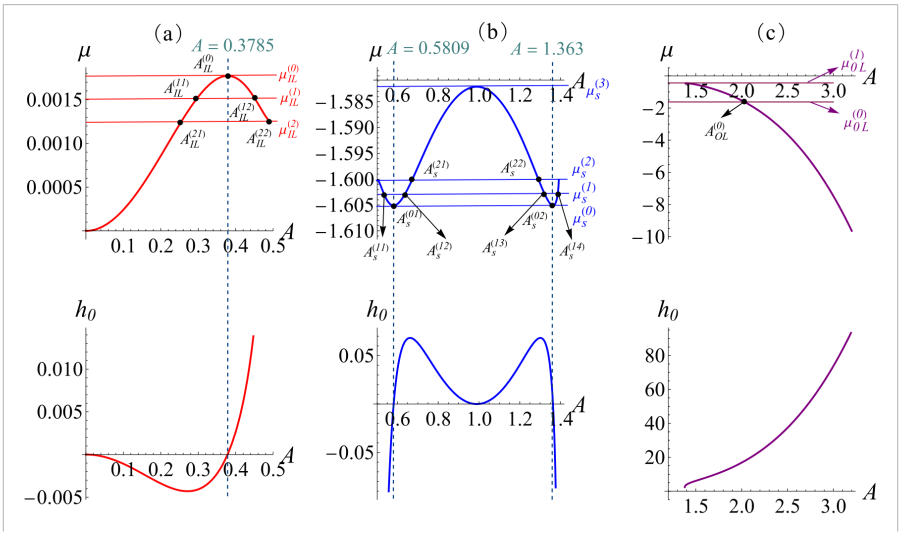

Figure 4: **Figure 6.** Relationship between the amplitude A and control parameter μ_c (upper), and the characteristic quantity h_0 (lower), for the triple-well oscillator. The evolution of both small limit cycles (within homoclinic orbits) and large limit cycles (enclosing all equilibria) is quantitatively captured.

7 Key Findings

1. **Full quantitative prediction:** The MGHFP method achieves, for the first time, analytical quantitative analysis of limit cycles in a coupled SD oscillator with two irrational nonlinearities. Given system parameters, one can directly compute the number, amplitude, and existence interval of every limit cycle—no repetitive numerical search needed.

2. **Analytical stability criterion:** The characteristic quantity h_0 distinguishes stable, unstable, and semi-stable limit cycles analytically. Unstable and semi-stable limit cycles—exceedingly difficult to capture numerically—are reliably identified by this method.

3. **Simultaneous homo-heteroclinic bifurcation prediction:** For the triple-well system, homoclinic and heteroclinic orbits coexist at the same saddle. The MGHFP method predicts both bifurcation parameters simultaneously and provides analytical orbit solutions (Figures 10–11), a capability not available from prior approaches.

4. **Complete limit cycle lifecycle:** The entire lifecycle is quantitatively tracked—from birth (semi-stable bifurcation at $\mu^{(0)}$), through evolution (splitting into stable + unstable pair), to annihilation (collapse to singular point at $\mu^{(2)}$), as clearly shown in Figure 1.

5. **High-order damping applicability:** With quartic nonlinear damping, the method maintains reliable accuracy when $|\varepsilon\mu_i| \leq 0.5$. Beyond this range, accuracy is recovered by increasing the Fourier truncation order.

8 Method Limitations

- *Large perturbation parameter* ($\varepsilon > 0.5$): Relative error exceeds 5%. Higher-order perturbation expansions can mitigate this.

- *Limit cycles crossing saddle points*: Velocity drops sharply near the saddle, causing abrupt waveform features. Higher Fourier truncation orders are needed.

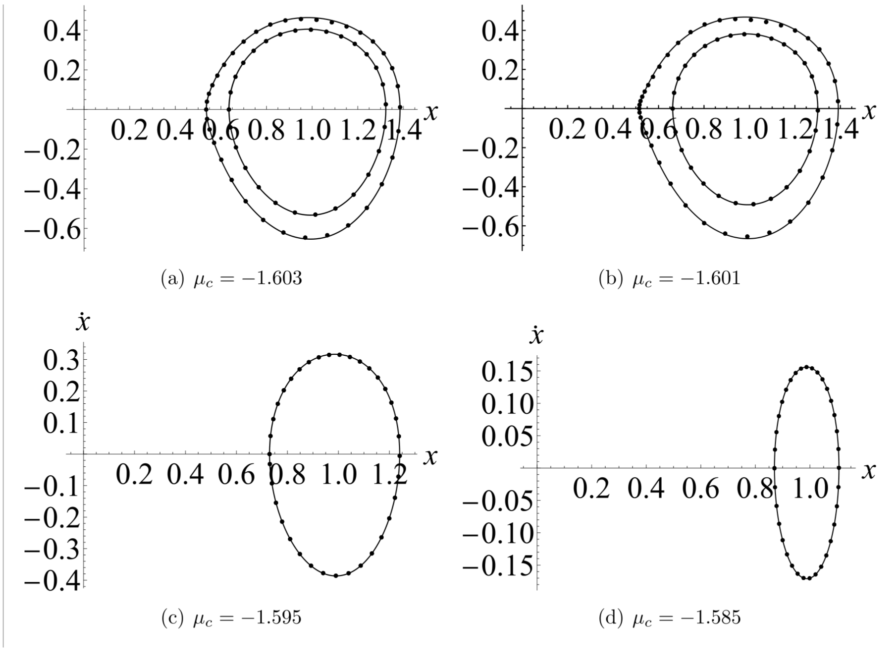

Figure 5: **Figure 7.** Limit cycles for the triple-well oscillator at different μ_c values ($\varepsilon = 1$). Solid lines: Runge–Kutta method. Dotted lines: MGHFP method. Despite the large perturbation parameter, the analytical results closely match the numerical reference.

- *Non-smooth potential wells* ($\alpha = 0$): Non-smoothness affects accuracy. A smaller ε or higher truncation order is recommended.

9 Applications

The model—combining irrational nonlinear stiffness, high-order nonlinear damping, and multi-stable dynamics—has direct relevance to:

(1) QZS vibration isolators: Quasi-zero stiffness isolators with nonlinear electromagnetic damping share the irrational stiffness and high-order damping structure. The MGHFP method provides a quantitative tool for predicting their limit cycle behavior and bifurcation thresholds.

(2) MEMS resonators: In micro-electro-mechanical systems, nonlinear damping and multi-stable dynamics are increasingly exploited. The bifurcation analysis framework directly supports design and optimization where saddle-node bifurcations and multi-stability play critical roles.

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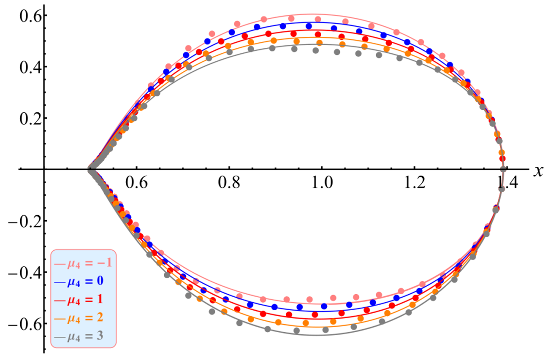
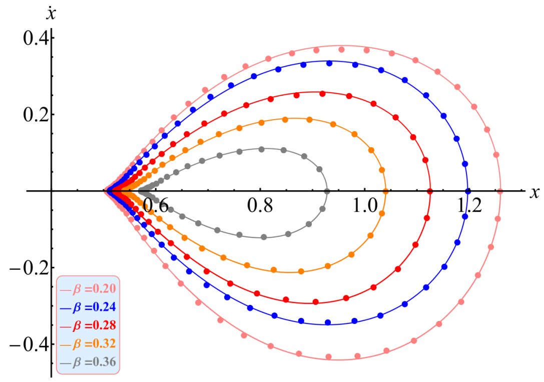
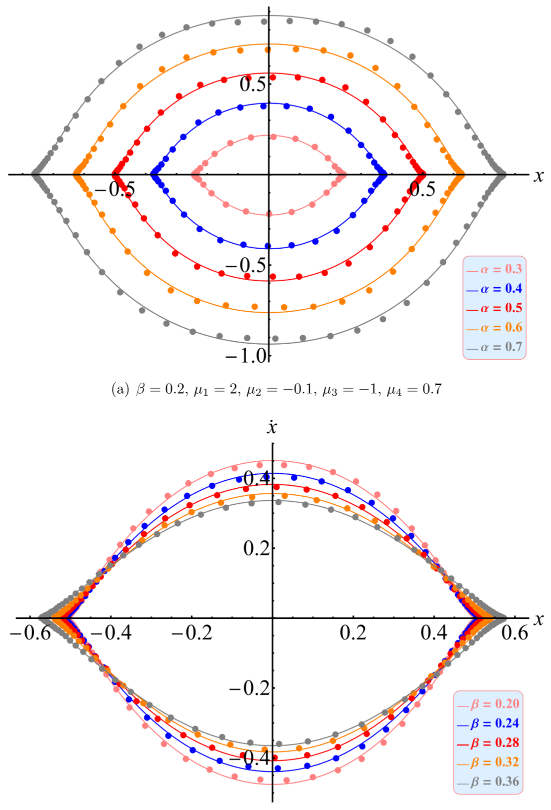


(a) $\alpha = 0.5$, $\beta = 0.2$, $\mu_2 = -0.1$, $\mu_3 = -1$, $\mu_4 = 0.7$



(b) $\alpha = 0.5$, $\mu_1 = 2$, $\mu_2 = -0.1$, $\mu_3 = -1$, $\mu_4 = 0.7$

Figure 6: **Figure 10.** Phase portraits of homoclinic solutions with different parameters ($\varepsilon = 0.5$). Solid lines: Runge–Kutta. Dotted: MGHFP. The method accurately captures the homoclinic orbit geometry even at relatively large perturbation.



(a) $\beta = 0.2$, $\mu_1 = 2$, $\mu_2 = -0.1$, $\mu_3 = -1$, $\mu_4 = 0.7$

(b) $\alpha = 0.5$, $\mu_1 = 2$, $\mu_2 = -0.1$, $\mu_3 = -1$, $\mu_4 = 0.7$

Figure 7: **Figure 11.** Phase portraits of heteroclinic solutions with different parameters ($\varepsilon = 0.1$). Solid lines: Runge–Kutta. Dotted: MGHFP. The analytical heteroclinic orbits connecting two saddle points are in excellent agreement with numerical results.